Quicksort

Topic: DivideAndConquer File: d2.pdf

PROBLEM

Sort an array by selecting a pivot and partitioning values around it.

ALGORITHM

Partition so values smaller than the pivot go left and larger values go right; recursively sort the two partitions.

EXAMPLE

For [5, 2, 8, 1, 6], pivot 5 can partition smaller values [2,1] from larger values [8,6], then sort each side.

COMPLEXITY

Average and expected time with good/random pivots: O(n log n). Worst case: O(n^2), for highly unbalanced partitions.

SPACE

Average recursion stack is O(log n); worst-case stack can be O(n). In-place partitioning uses little extra array storage.

TIP

Randomized pivot selection reduces the chance of repeatedly choosing a poor pivot.